Automating Antimicrobial Resistance Genome Analysis Through a Bioinformatic Pipeline

Kenneth Tran[1,2], ShuTing Chen[2], Dr. Xu Yang[2]
[1] West Covina High School, [2] Nutrition and Food Science Department, Cal Poly Pomona

Cal Poly Pomona

Abstract

The prevalence of antibiotic-resistant microbes outside of hospital settings has increased rapidly, leading researchers to study the dissemination of antibiotic-resistant bacteria in the food chain. Numerous bioinformatics applications exist for genome assembly, resistance gene identification, sequence typing, and mobile genetic element analysis. Still, researchers typically use these tools in a series of fragmented steps, leading to unnecessary labor and time, and potential errors due to misorganization. To address these limitations, we developed an integrated bioinformatics pipeline that consolidates these analyses into a single, streamlined workflow for the genomic characterization of antibiotic-resistant bacteria. The pipeline automates genome assembly and quality assessment, antimicrobial resistance gene identification, sequence typing, and mobile genetic element analysis while generating standardized and organized outputs. By reducing manual data handling and integrating commonly used genomic analyses, this workflow improves the reproducibility and accessibility of bacterial genome analysis.

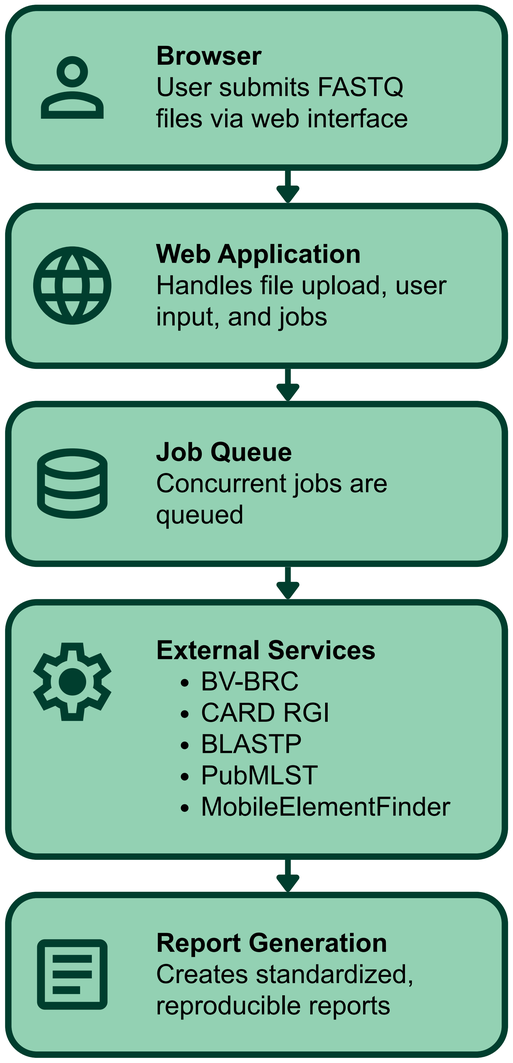
Introduction

- Whole-genome sequencing is now essential for antimicrobial resistance (AMR) surveillance
- Existing tools cover genome assembly, resistance gene ID, sequence typing, and mobile genetic element analysis, but are usually run independently
- Running tools independently requires repeated data transfers and manual result aggregation; labor-intensive and error-prone
- We present a web-based pipeline automating the full AMR workflow, from paired Illumina FASTQ files through final report generation
- Enables analysis of bacterial isolates with minimal manual intervention

Methodology

- Databases (CARD RGI, BLASTP, PubMLST, MobileElementFinder) are kept locally for efficiency, at the cost of weekly refreshes
- BLASTP database is narrow: NCBI's full database (733 GB) is too large to host locally, and remote submissions timed out (1800s) without returning results.
- Used NCBI's curated AMRFinderPlus protein catalogue instead (~5 MB, results in seconds); remote search retained only as fallback for enzymes AMRFinderPlus can't name
- Refreshes scheduled Sundays at 03:00 local time to avoid disrupting peak upload windows
- Interpretation rules were constrained:
  - Resistance calls collapsed to one gene per predicted sequence (not one per CARD variant)
  - Loose RGI hits excluded from mobile-element linkage
  - Colocation asserted only within 5000 bp on the same contig

Methodology (cont.)



Future Work

Future work will focus on improving scalability and usability. One enhancement will be the introduction of customizable, modular workflows, allowing users to enable, disable, or replace analysis tools to accommodate different research objectives and computational resources. Integration of additional resistance databases and machine learning approaches could further enhance the identification of novel resistance determinants.

Results

- Manual workflow required 12 hands-on interventions for a single sample, increasing linearly with batch size (slope ≈ 0.8, i.e., +4 interventions per +5 samples)
- The pipeline required 1 intervention/sample regardless of batch size.
- Centralized, standardized outputs from pipeline versus scattered outputs during manual work

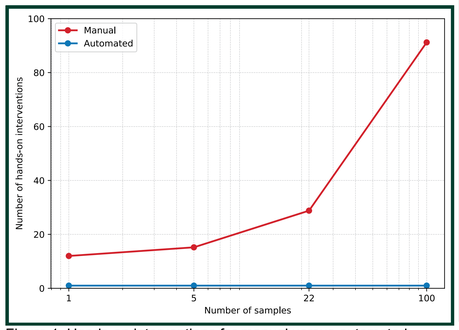


Figure 1. Hands-on interventions for manual versus automated analysis across increasing batch sizes.

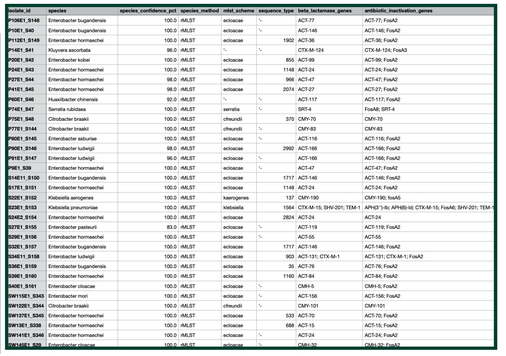


Figure 3. Master report generated by the pipeline.

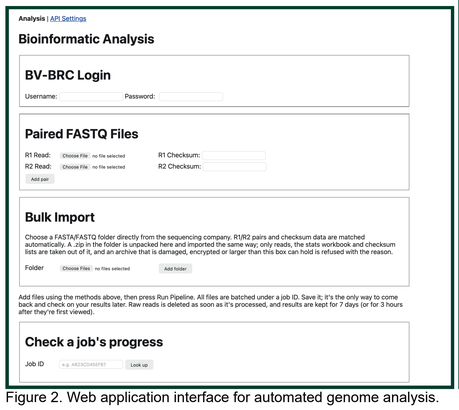


Figure 2. Web application interface for automated genome analysis.

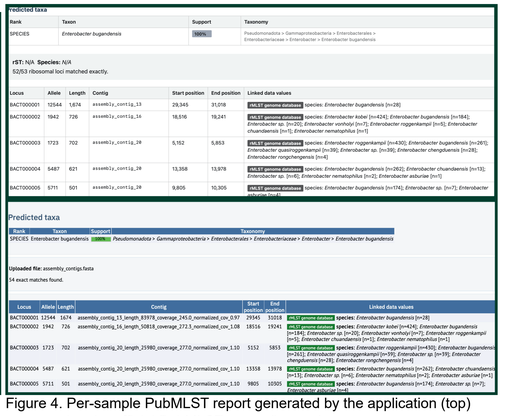


Figure 4. Per-sample PubMLST report generated by the application (top) compared to the official PubMLST report (bottom).

Conclusion

The presented pipeline integrates multiple well-established bioinformatics tools into an automated workflow for antimicrobial resistance surveillance. By automating genome assembly, resistance identification, sequence typing, mobile element detection, and report generation from FASTQ sequencing data, the system minimized manual effort while supporting scalable analysis of bacterial isolate collections from food samples.

Acknowledgements
We thank the USDA for funding this project and Doanho Ha for her support and training throughout the project.

References & Additional Information
Scan for references and additional information.